Retail Analytics: Forecasting Challenge

Project Requirements

Objective

The objective is for you to experience a pseudo-real-life demand forecasting task in the context of online retailing via the interactive case study. To complete the task, you will need to go through a complete cycle of developing and refining forecasting algorithm solutions.

Scope

The project is group based. We focus on the scenario "test_world" of the Retail Analytics case study (each group has been granted **read** access to all the data and also access to upload forecast). The scenario is self-contained, so no external data is needed or allowed.

Task Description

The "test_world" scenario will evolve based on its own internal logic of customer arrival, recommendation algorithm, pricing strategy, and inventory replenishment policy. The participants' task is to observe the historical data and forecast the option sales in the future. Note that the uploaded forecast will not be used in the inventory planning and hence will not affect how the world evolves.

Forecasting specifications:

1. Target: option sales quantity.
2. Target granularity: option * day.
3. Forecast horizon: 28 days.
4. Execution frequency: weekly.
5. Accuracy metric: 1-MAPE accuracy weighted by the actual sales quantity.
6. Time scope:

The following four rounds of forecasting will be evaluated altogether according to the accuracy metric defined above.

Round ID	Forecast Date	Feature Date	Target Date Range
1	20220916	20220915	20220916—20221013
2	20220923	20220922	20220923—20221020
3	20220930	20220929	20220930—20221027

Round ID	Forecast Date	Feature Date	Target Date Range
4	20221007	20221006	20221007—20221103

	20220916 20220922	20220923 20220929	20220930 20221006	20221007 20221013	20221014 20221020	20221021 20221027	20221028 20221103
feature_date = 20220915							
feature_date = 20220922							
feature_date = 20220929							
feature_date = 20221006							

Submission Deadlines

As the "test_world" need to continue to evolve and generate new data, the participants must pay attention to the following submission deadlines. Late submission is not possible.

1. By **April 13 (Monday)** 23:59:59. This is the testing period. "test_world" will pause at 20220715, and participants can test the APIs for uploading forecast. After the deadline, the world will move on to 20220916, and data up to 20220915 will be available.

2. By **April 16 (Thursday)** 23:59:59. Submit the forecast of Round 1 before the deadline. During the period between April 14~16, you can upload multiple times and the new ones will overwrite the old ones. After the deadline, the world will move on to 20220923, and data up to 20220922 will be available. Forecast for Round 1 cannot be uploaded/updated any more.

3. By **April 19 (Sunday)** 23:59:59. Submit for Round 2. The world will move on to 20220930 after the deadline.

4. By **April 22 (Wednesday)** 23:59:59. Submit for Round 3. The world will move on to 20221007 after the deadline.

5. By **April 25 (Saturday)** 23:59:59. Submit for Round 4. The world will move on to 20221103 after the deadline, and final accuracy evaluation will be conducted.

6. By **May 3 (Sunday)** 23:59:59. Submit your report for the competition on Canvas. Describe your forecasting algorithm solutions. Page limit: 5.

Notes:

1. Submission is group based. Every player can submit on behalf of the group. Later submission will overwrite earlier submission of the group.

2. It is the participant's responsibility to make sure there is no missing forecast. Missing forecast will be replaced with zeros.

Grading Criteria

1. Overall accuracy (70%).

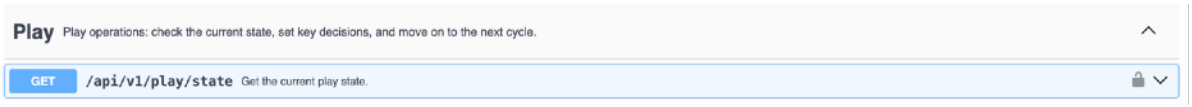
2. Report (30%).

Submission Instructions

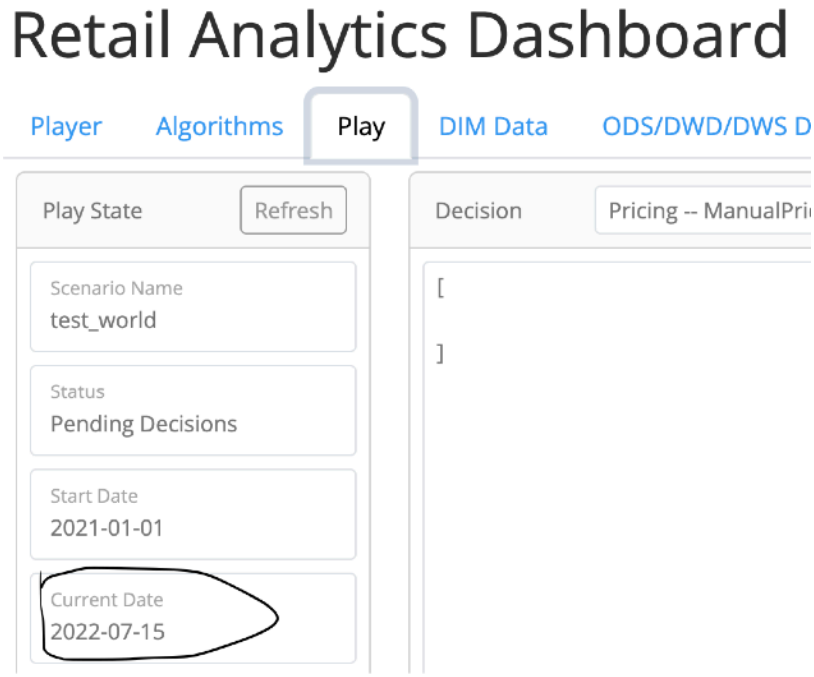
Below are the relevant APIs and dashboard functions.

1. Check the state of the world, especially the current date of the world.

Using API:



Using the dashboard:



2. Submit forecast.

Use the following APIs:

Competition Forecasting Competition.

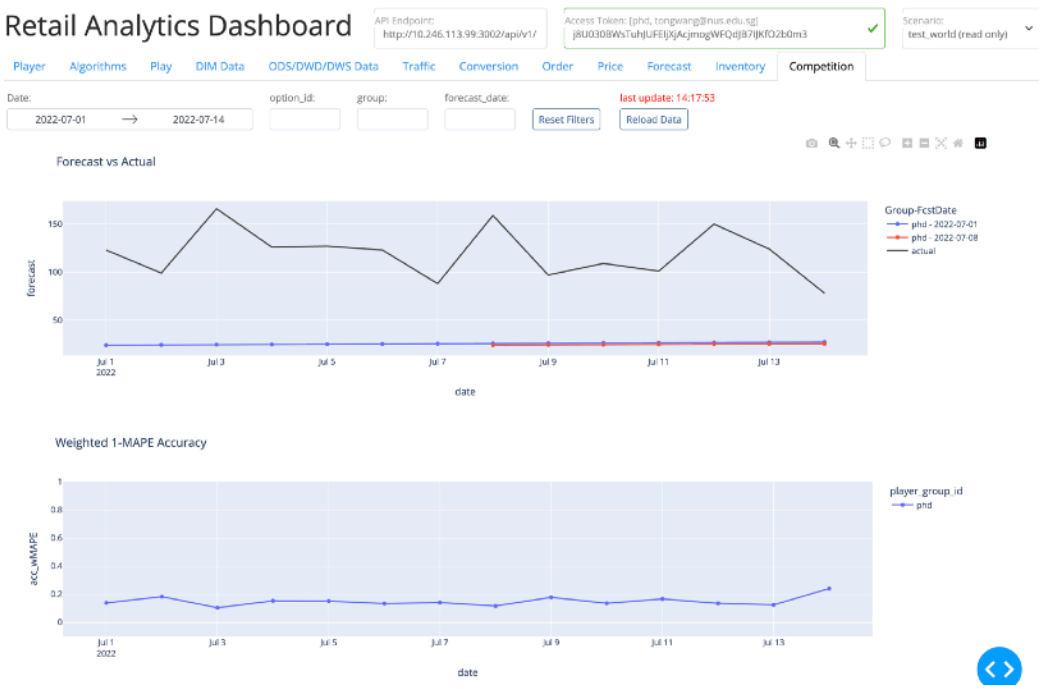
GET	/api/v1/competition/forecast_competition_template	Get the forecast template for the competition.
GET	/api/v1/competition/forecast_competition_input	GET forecast for the competition.
POST	/api/v1/competition/forecast_competition_input	POST forecast for the competition.

a. Get the template: GET *competition/forecast_competition_template*. The template provide the expected format for submitting the data. It is a list of dict with keys "option_id" and "forecast," where the "forecast" is expected to be a list of 28 float numbers indicating the forecast of the next 28 days. You can use the template and replace the numbers with your forecast.

b. Upload forecast: POST *competition/forecast_competition_input*. Use the POST api to upload the forecast data already in the expected format. Check http://10.246.113.99:3002/public/api_access.html for how to post data using Python.

NOTE: the data POSTed to the API must strictly follow the JSON syntax (https://www.w3schools.com/js/js_json_syntax.asp), otherwise the API may fail because of failure in parsing the data.

c. Double check the submission: GET *competition/forecast_competition_input*. Use the GET api to double check whether the uploaded forecast is as expected.

3. Check the performance.

Use the dashboard's "Competition" tab for intermediate competition results. The result is available only after the world has proceeded to the next week and the submission is closed.